8. Using factor theorem, show that $(x-5)$ is a factor of the polynomial $2x^3-5x^2-28x+15$

9. Determine the value of m, if $(x+3)$ is a factor of $x^3-3x^2-mx+24$.

10. If both $(x-2)$ and $\left(x-\frac{1}{2}\right)$ are the factors of ax^2+5x+b, then show that $a=b$.

11. If $(x-1)$ divides the polynomial $kx^3-2x^2+25x-26$ without remainder, then find the value of k.

12. Check if $(x+2)$ and $(x-4)$ are the sides of a rectangle whose area is x^2-2x-8 by using factor theorem.

3.4 Algebraic Identities

An identity is an equality that remains true regardless of the values chosen for its variables.

We have already learnt about the following identities:

(1) $(a+b)^2 \equiv a^2+2ab+b^2$ (2) $(a-b)^2 \equiv a^2-2ab+b^2$

(3) $(a+b)(a-b) \equiv a^2-b^2$ (4) $(x+a)(x+b) \equiv x^2+(a+b)x+ab$

Note

(i) $a^2+b^2=(a+b)^2-2ab$ (ii) $a^2+b^2=(a-b)^2+2ab$

(iii) $a^2+\frac{1}{a^2}=\left(a+\frac{1}{a}\right)^2-2$ (iv) $a^2+\frac{1}{a^2}=\left(a-\frac{1}{a}\right)^2+2$

Example 3.16 Expand the following using identities: (i) $(3x+4y)^2$ (ii) $(2a-3b)^2$ (iii) $(5x+4y)(5x-4y)$ (iv) $(m+5)(m-8)$

Solution

(i) $(3x+4y)^2$ $\left[\text{we have } (a+b)^2=a^2+2ab+b^2\right]$

$(3x+4y)^2=(3x)^2+2(3x)(4y)+(4y)^2$ put $[a=3x,\ b=4y]$

$=9x^2+24xy+16y^2$

(ii) $(2a-3b)^2$ $\left[\text{we have } (a-b)^2=a^2-2ab+b^2\right]$

$(2a-3b)^2=(2a)^2-2(2a)(3b)+(3b)^2$ put $[a=2a,\ b=3b]$

$=4a^2-12ab+9b^2$

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